

13	A 350 cm ³ cup of hot tea takes 30 minutes to cool down from an initial temperature of 90 °C to 30 °C.							
	The specific heat capacity of tea is 4200 J kg ⁻¹ K ⁻¹ and its density is 1000 kg m ⁻³ .							
	What is the average rate of heat loss of the tea?							
	A	49 W	B	2900 W	C	4.9 × 10 ⁴ W	D	2.9 × 10 ⁶ W
	Answer: A							
	Rate of heat loss = $\frac{Q}{t} = \frac{mc\Delta T}{t} = \frac{(0.350)(4200)(90 - 30)}{(30)(60)} = 49 \text{ W}$							

14	An object is moving in a horizontal simple harmonic motion parallel to the edge of a metre rule.
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